

Superposition Theorem

Actual Statement of Theorem

The Superposition Theorem states that in any linear electrical circuit containing multiple independent sources, the voltage or current in any element is equal to the algebraic sum of the voltages or currents produced by each independent source acting alone.

While considering one source at a time:

- All other voltage sources are replaced by short circuits
- All other current sources are replaced by open circuits

$$\text{Total Response} = V_1 + V_2 + V_3 + \dots$$

This principle allows complex circuits to be broken down into simpler parts, making analysis easier.

Requirement: Linear Circuits

Superposition is only valid for **linear circuits**. A linear circuit is one in which the relationship between voltage and current is proportional and follows Ohm's Law.

$$V = IR$$

In linear systems:

- Doubling the input doubles the output
- Circuit parameters remain constant
- Responses can be added algebraically

Non-linear components such as diodes and transistors violate this property and cannot be analyzed using superposition.

Step-by-Step Procedure

Step 1: Identify All Independent Sources

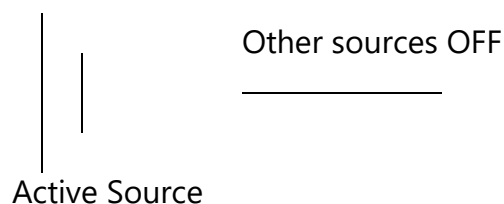
Locate all independent voltage and current sources present in the circuit. Each source will be analyzed separately.

Step 2: Activate One Source at a Time

Select one source to remain active and deactivate all others:

- Voltage sources → Short circuit
- Current sources → Open circuit

This isolates the effect of a single source.



Step 3: Calculate Individual Contribution

Using circuit analysis methods, determine the voltage or current contribution from the active source.

$I_1, I_2, I_3 \dots$

Step 4: Repeat for All Sources and Add Results

Repeat the process for each source and then combine all contributions algebraically.

$$I_{\text{total}} = I_1 + I_2 + I_3$$

Signs must be carefully considered while adding results.

Methods to Solve Each Step

1. Nodal Analysis

Applies Kirchhoff's Current Law at nodes to determine voltages.

Use when: Circuit has many nodes and parallel paths.

2. Mesh Analysis

Applies Kirchhoff's Voltage Law in loops to determine currents.

Use when: Circuit has fewer loops.

3. Ohm's Law

Used for simple circuits to directly calculate current or voltage.

$$I = V / R$$

Use when: Circuit is already simple.

4. Source Transformation

Convert voltage sources to current sources and vice versa.

Use when: Simplifying mixed circuits.

5. Voltage Divider Rule

Used to calculate voltage in series circuits.

6. Current Divider Rule

Used in parallel circuits to divide current.

Choosing the correct method depends on circuit structure. Efficient selection reduces calculation complexity.

Important Notes

Superposition can only be applied to voltages and currents. Power cannot be directly calculated using superposition because it is a non-linear function.

Each source must be considered independently, and final results must include proper sign conventions.

Practical Applications

Superposition is widely used in electrical engineering for analyzing circuits with multiple power sources. In power systems, it helps determine the contribution of each generator to the total output.

In electronics, it is used to analyze circuits with multiple signals, such as amplifiers and communication systems.

It is also useful in troubleshooting, where engineers isolate the effect of individual sources to identify faults in a system.

Overall, the theorem provides a systematic approach to understanding complex circuits by breaking them into simpler components.